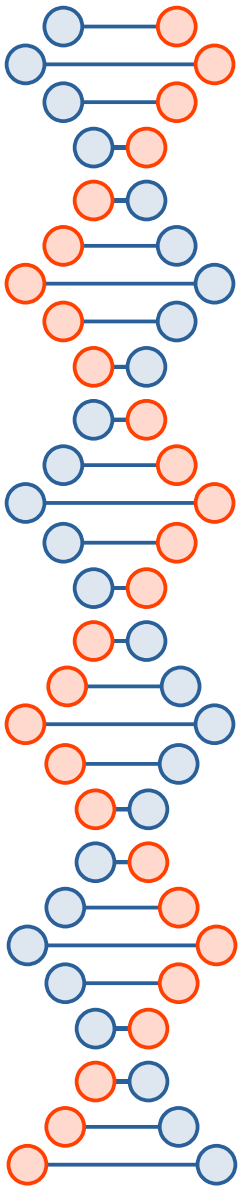


Tips to apply when **CONVERTING UNITS**

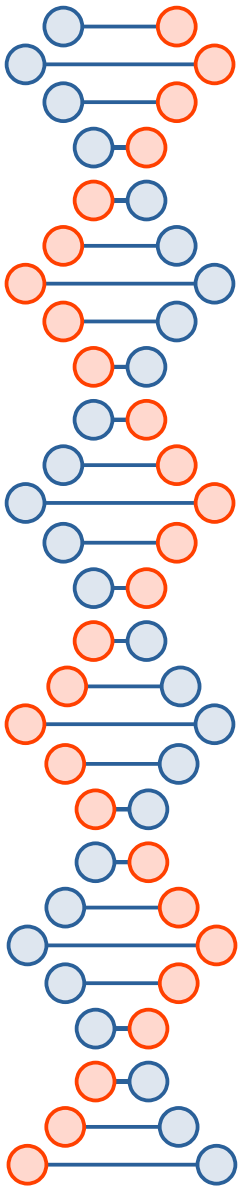
How to prevent making those errors
during units conversion?

Lubos Polerecky & Karline Soetaert



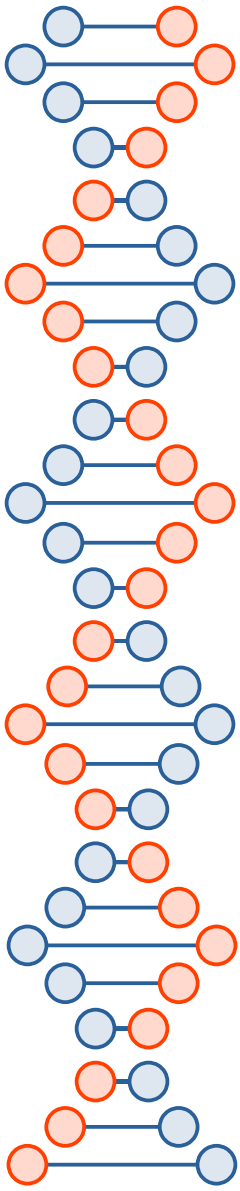
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- Units play an important role when solving quantitative problems.



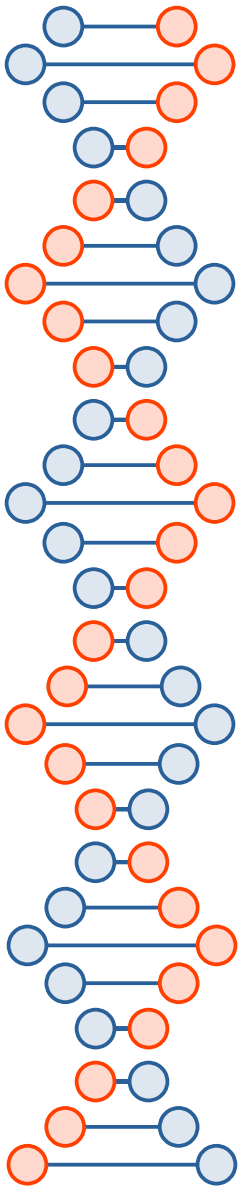
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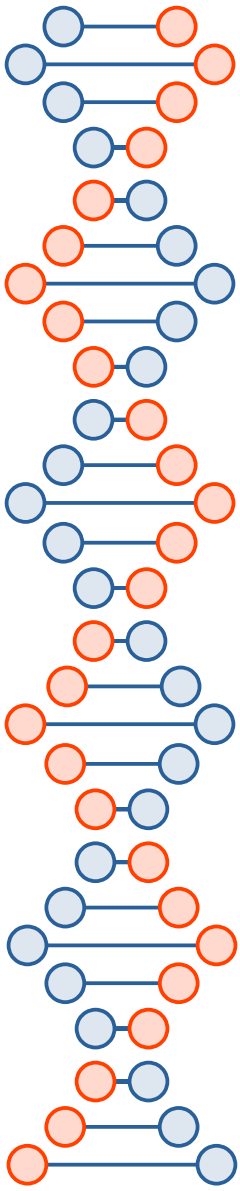
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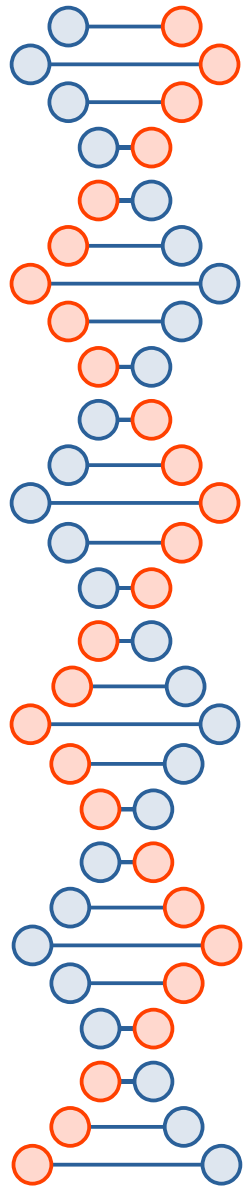
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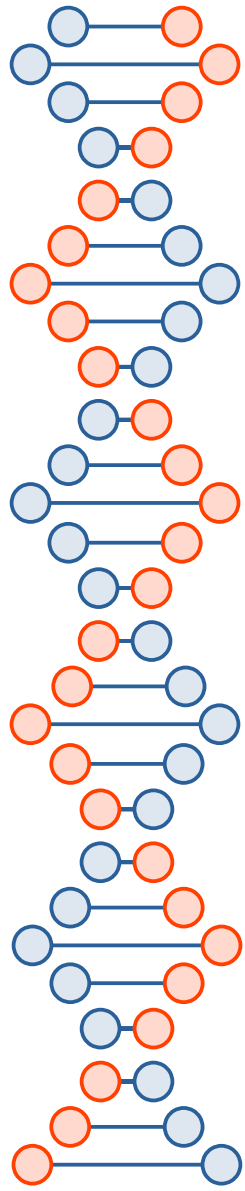
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 - the first **quality check** of our models will very often include a **check of units**; in most cases, it will also be the most effective one.
 - Our basic rationale is: *if the units are wrong, the model cannot be right.*



Our experience:

When solving quantitative problems, **most frequent mistakes** made by students occur **during units conversions**.

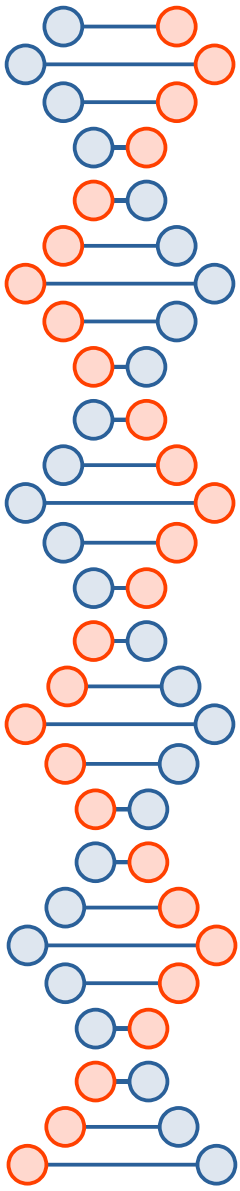


Our experience:

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Our aim here:

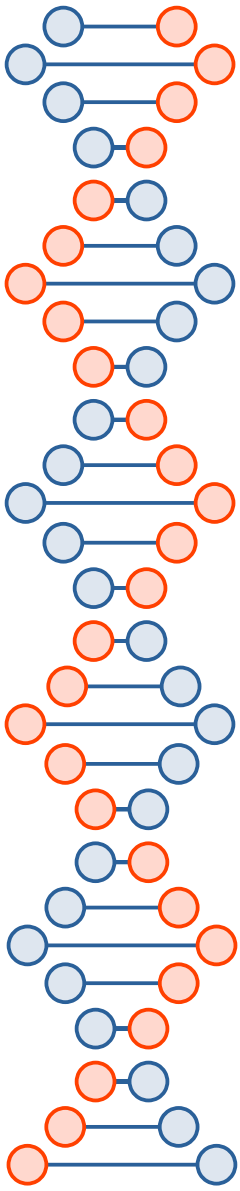
To give you tips on **how to avoid** making such mistakes.



Recommended steps:

Step 1:

Convert all prefixes to the form **10^n** , where **n** is an **integer** number.



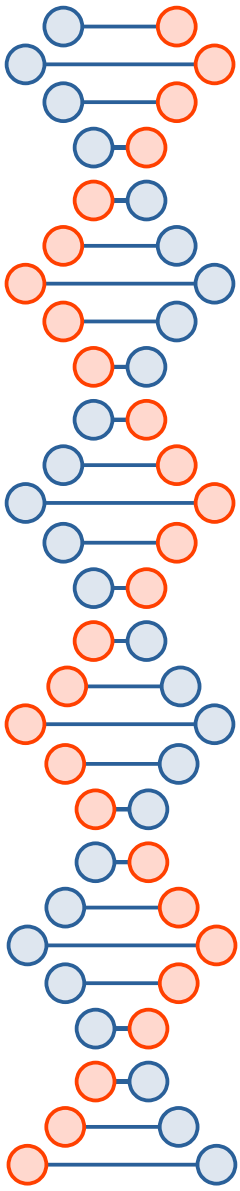
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Convert all units to **base SI units**.



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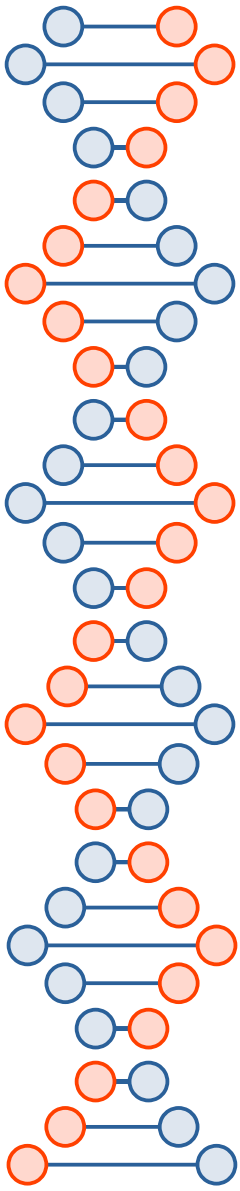
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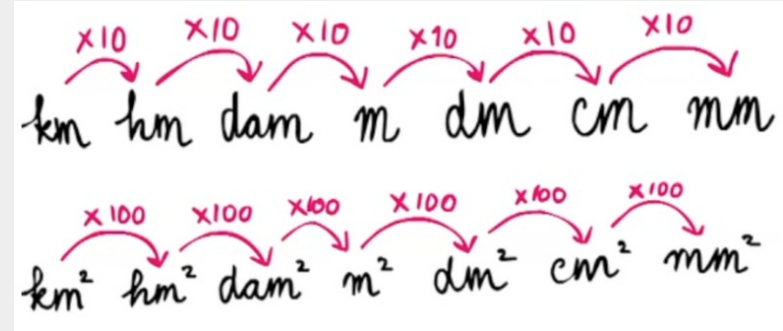
Simplify the expression with units using basic mathematical rules.

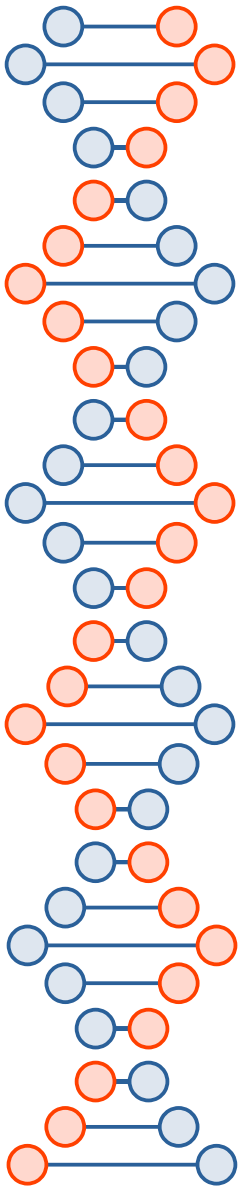


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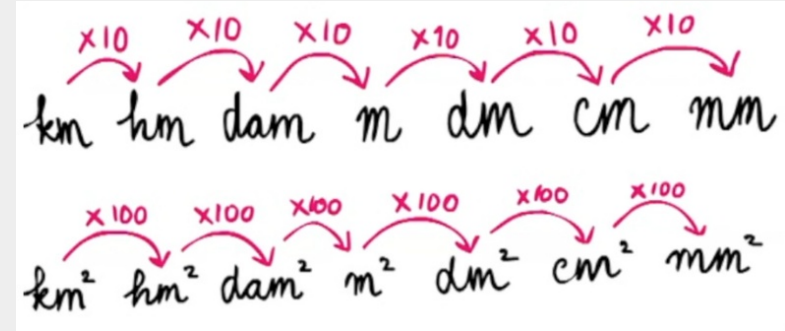




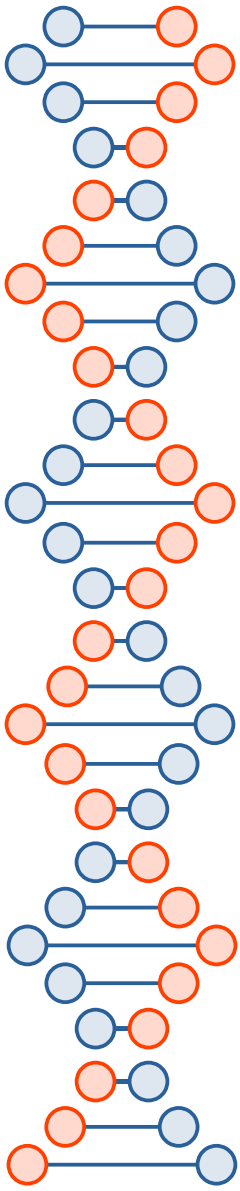
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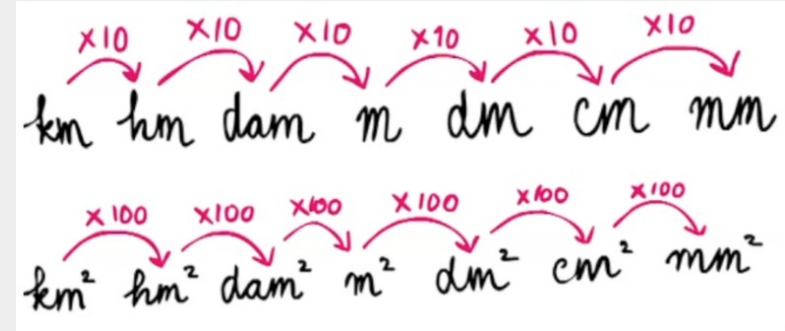
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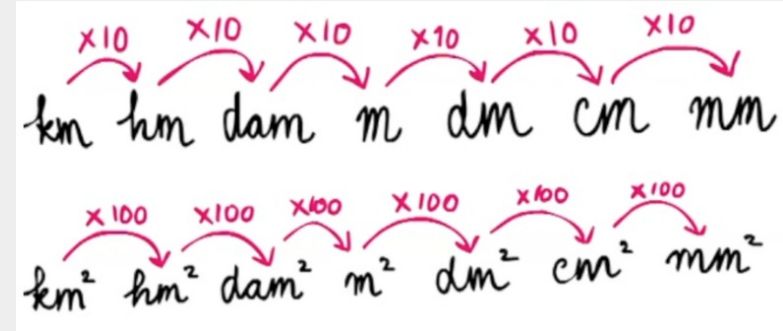


- they are only helpful in simple problems.
- In more complex problems, and more often than not, **they lead to confusion** and, ultimately, those **errors** that you want to avoid.

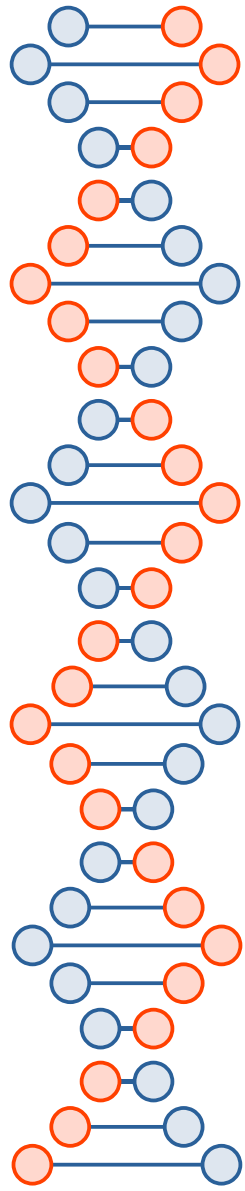
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- they are only helpful in simple problems.
- In more complex problems, and more often than not, **they lead to confusion** and, ultimately, those **errors** that you want to avoid.
- Your calculation **will be much less prone to errors** if you first convert all prefixes to the form **10^n** .



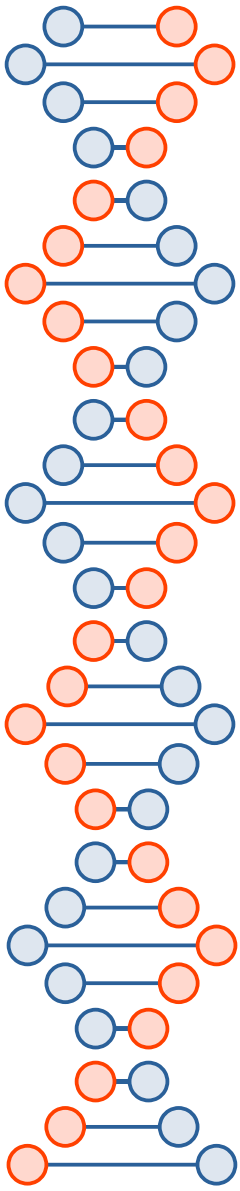
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Convert all prefixes to the form **10^n** , where n is an **integer** number.

The most common prefixes¹ and their meaning are summarized in this table:

Prefix name	Prefix symbol	10^n	Example
nano	n	10^{-9}	nm = 10^{-9} m
micro	μ	10^{-6}	μ s = 10^{-6} s
milli	m	10^{-3}	ms = 10^{-3} s
centi	c	10^{-2}	cm = 10^{-2} m
deci	d	10^{-1}	dm = 10^{-1} m
hecto	h	10^2	hPa = 10^2 Pa
kilo	k	10^3	km = 10^3 m
mega	M	10^6	MPa = 10^6 Pa

¹More prefixes can be found here: https://en.wikipedia.org/wiki/Metric_prefix



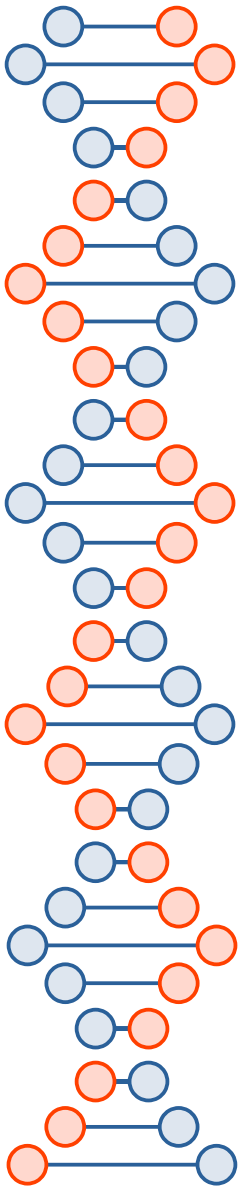
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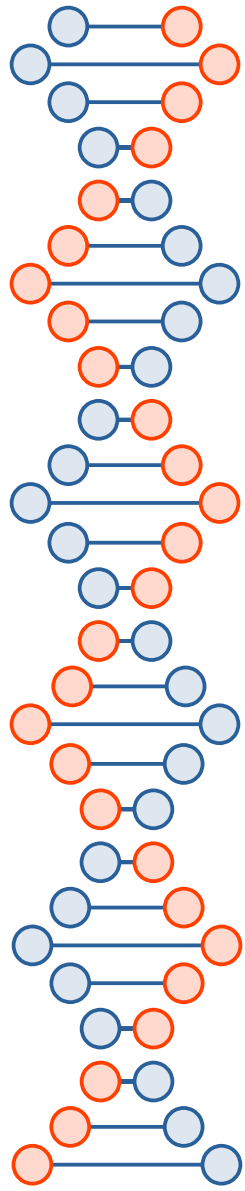


Step 2

Convert all units to **base SI units**.

Remember that these are the **base** SI units:

Unit	Name	Quantity
m	meter	length
s	second	time
kg	kilogram	mass
mol	mole	amount of substance
K	Kelvin	temperature
cd	candela	light
A	ampere	electric current



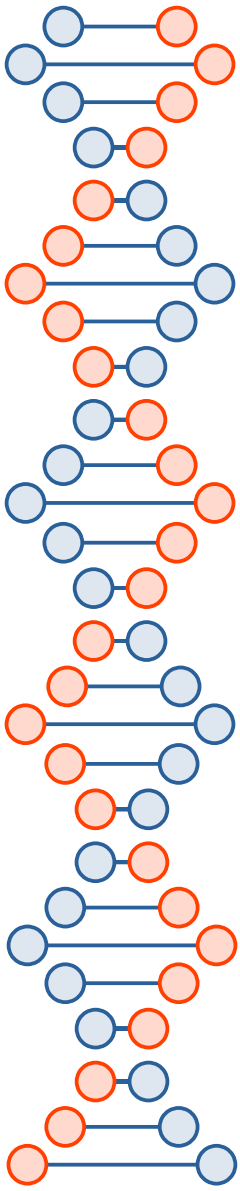
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Somewhat oddly,
kg is the base SI unit
of mass, **not gram (g)!!!**

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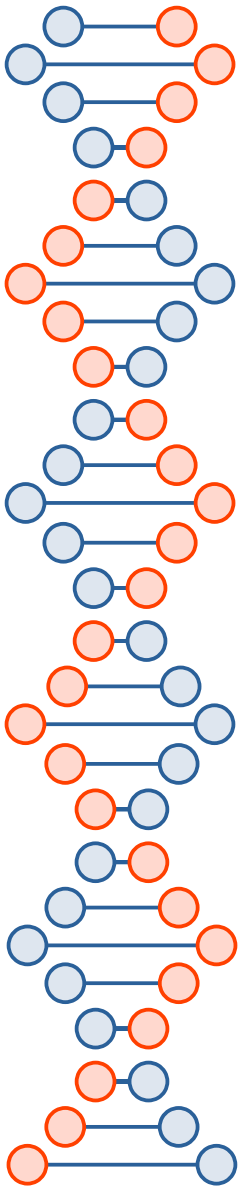
Convert all units to **base SI units**.

Every other unit is either

- **derived**

examples:

N	= Newton	= kg m s^{-2}
J	= Joule = N m	= $\text{kg m}^2 \text{s}^{-2}$
Pa	= Pascal = N m ⁻²	= $\text{kg m}^{-1} \text{s}^{-2}$



Step 2

Convert all units to **base SI units**.

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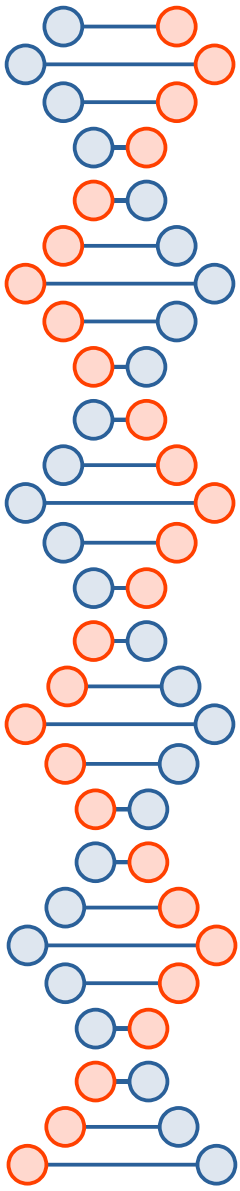
- or introduced for **convenience** or used for **“historical reasons”**

examples: L = liter = 1 dm^3

 min = minute = 60 s

 Atm = atmosphere = 101325 Pa

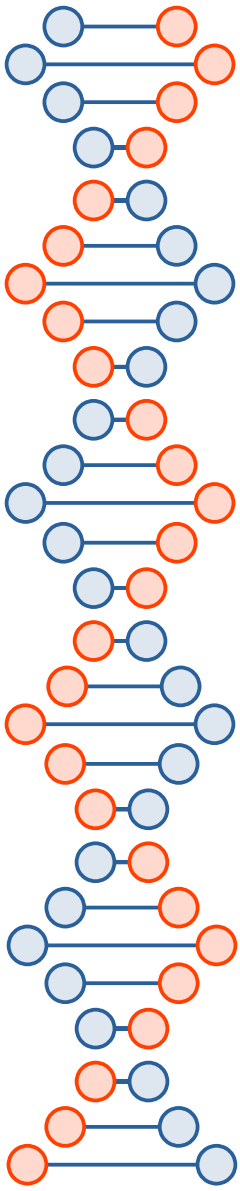
 M = molar = 1 mol L^{-1}



Step 2

Convert all units to **base SI units**.

We will, however, make **one exception**:



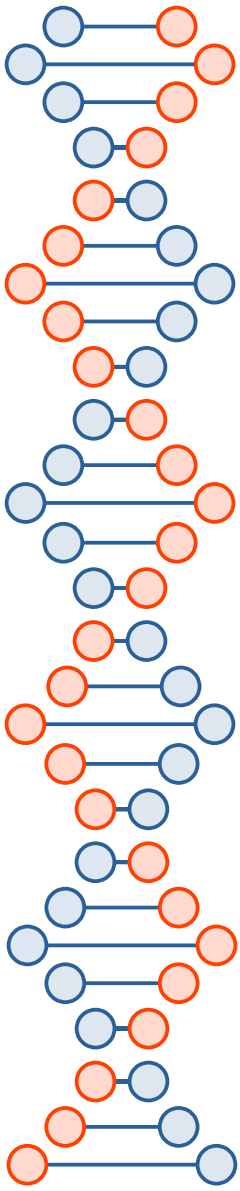
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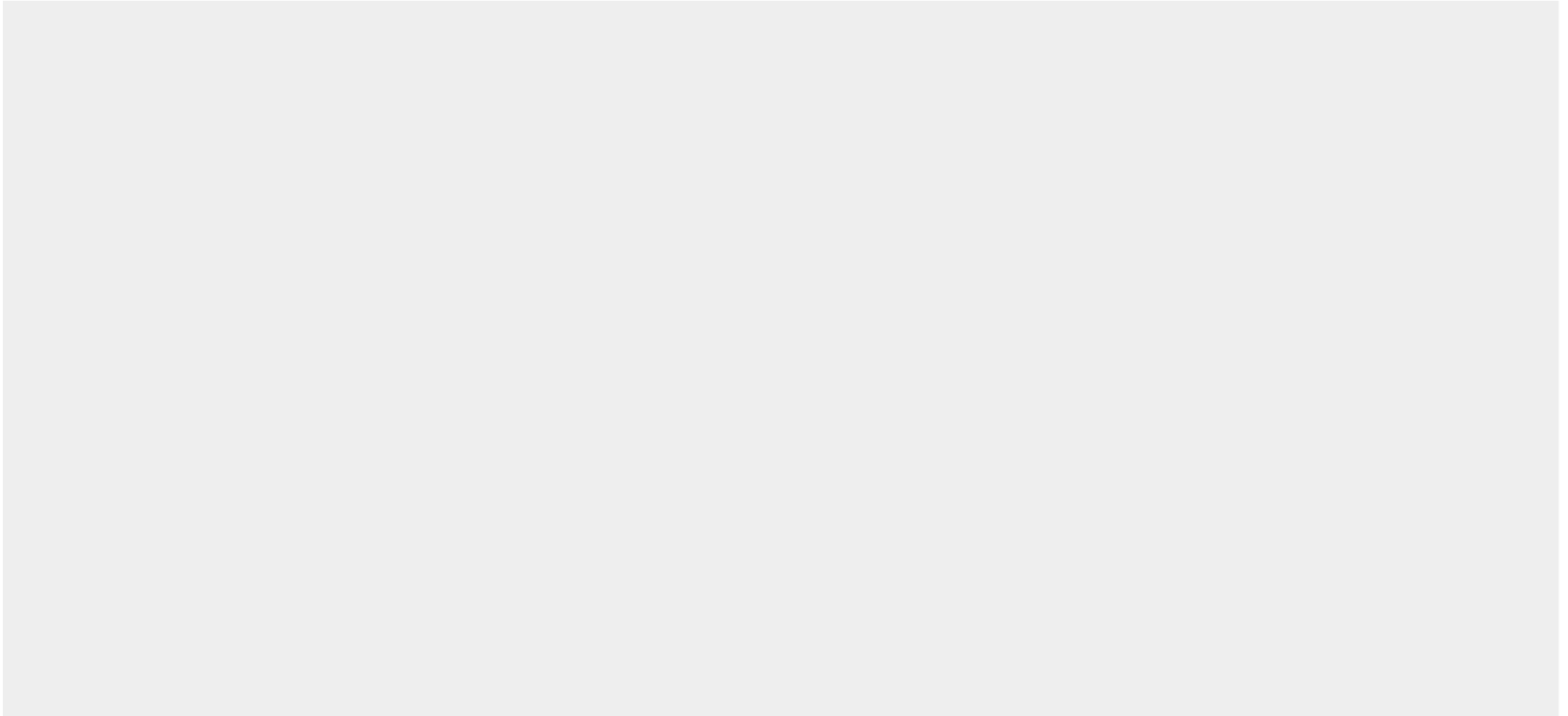
We will often be interested in results on **longer time scales** than seconds, for example over days or years.

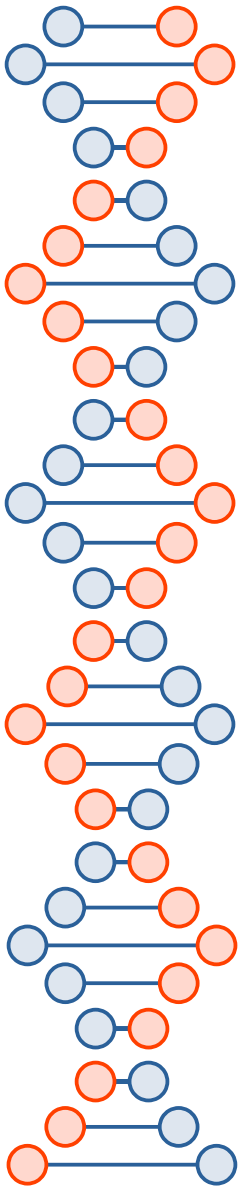
Thus, we will often use **days (d)** or **years (yr)** as an acceptable unit of time.



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Simplify the expression with units using basic mathematical rules:

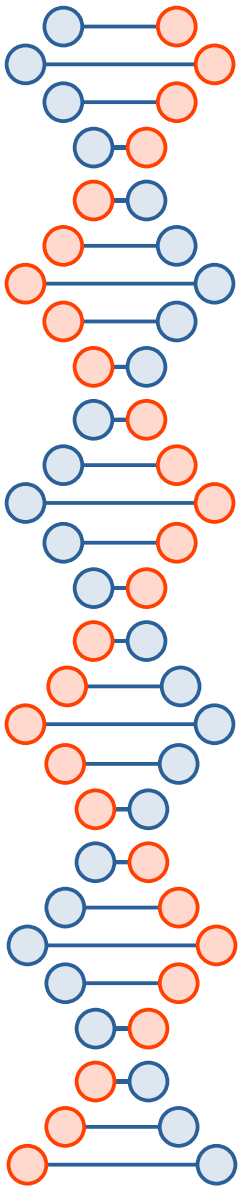




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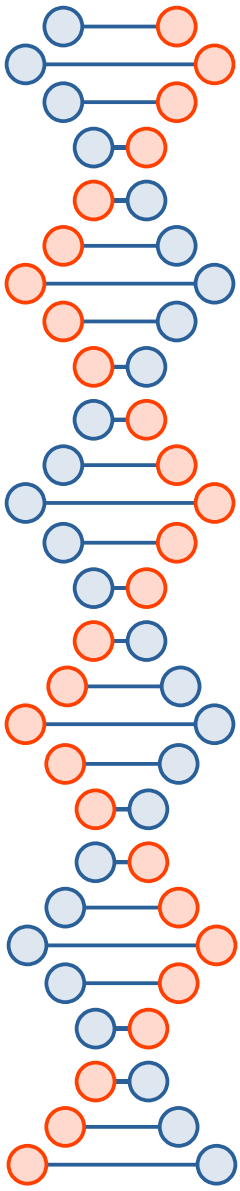
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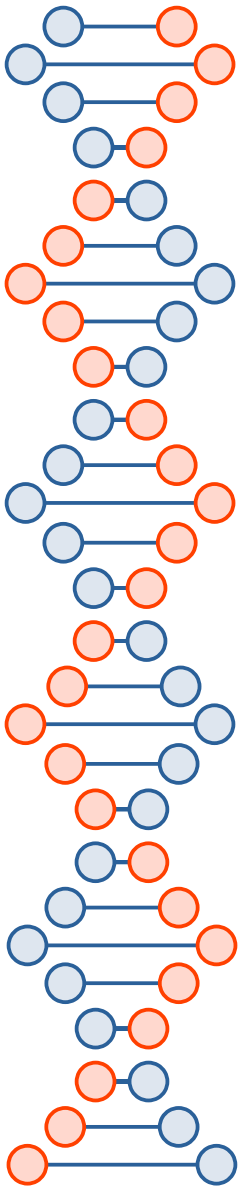
- First evaluate the **powers**
- Then continue with **multiplication** or **division** of units in the expression



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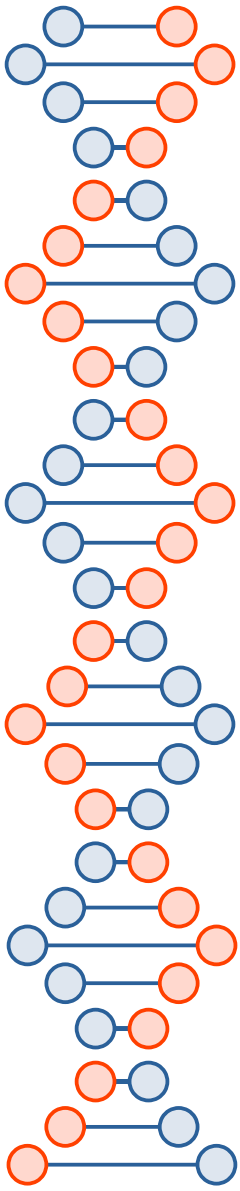
Simplify the expression with units using basic mathematical rules:

- First evaluate the **powers**
- Then continue with **multiplication** or **division** of units in the expression
- Only **add** or **subtract** quantities if they have the **same** unit.



Example 1

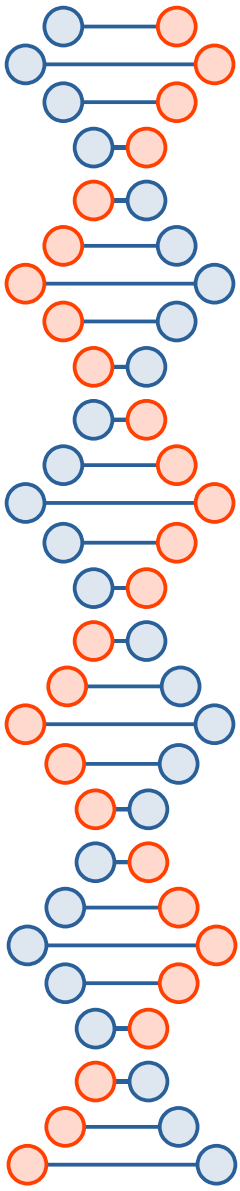
Convert the concentration of $200\ \mu\text{M}$ to mol m^{-3} .



Example 1

Convert the concentration of $200\text{ }\mu\text{M}$ to mol m^{-3} .

1. First, we find a relationship between L and m^3 : $\text{m}^3 = 1000\text{ L} = 10^3\text{ L}$ Therefore: $1\text{ L} = 10^{-3}\text{ m}^3$

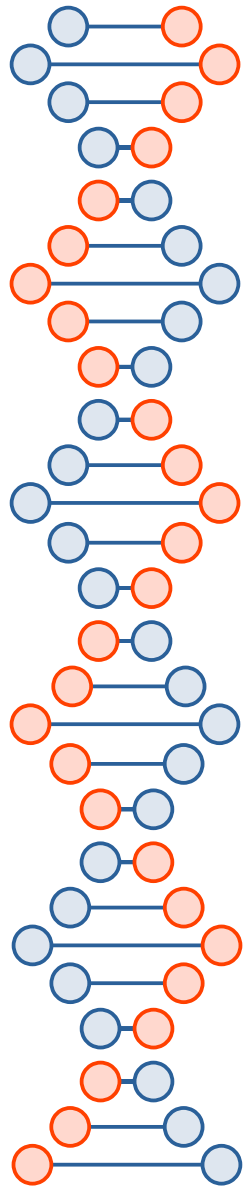


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$$200\ \mu\text{M} = 200 \times 10^{-6}\ \text{mol L}^{-1} = 200 \times 10^{-6}\ \frac{\text{mol}}{\text{L}}$$

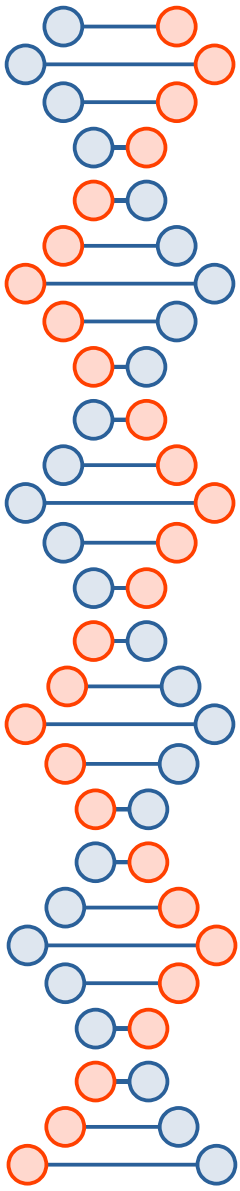


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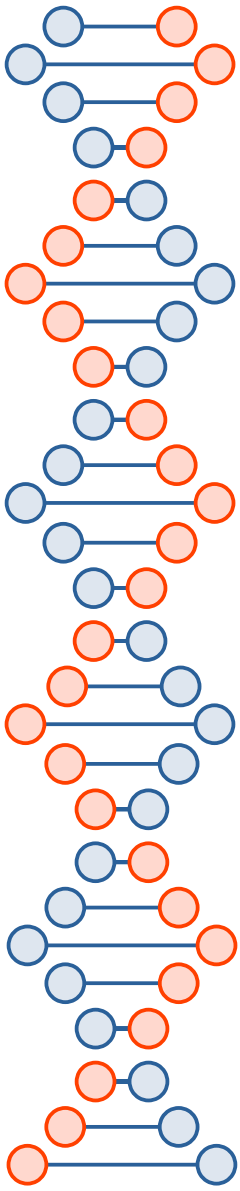
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Example 2

Convert the value of the diffusion coefficient $2 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$ to $\text{m}^2 \text{ yr}^{-1}$.

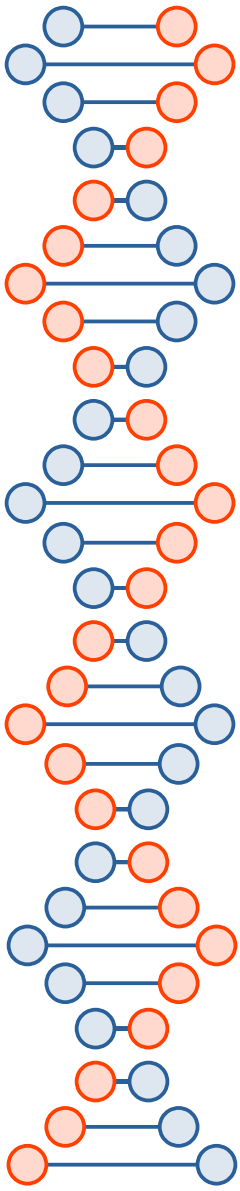


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$$\text{cm}^2 = (\text{cm})^2 = (10^{-2} \text{ m})^2 = (10^{-2})^2 \text{ m}^2 = 10^{-4} \text{ m}^2$$



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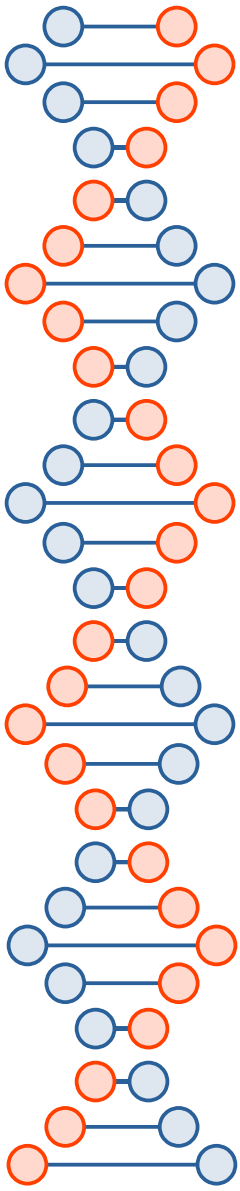
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$$\text{yr} = 365 \text{ d} = 365 \times 24 \times 60 \times 60 \text{ s} = 31,536,000 \text{ s} = 31.536 \times 10^6 \text{ s}$$



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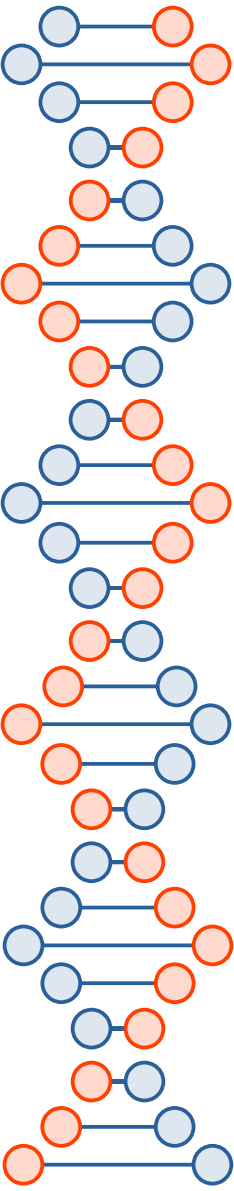
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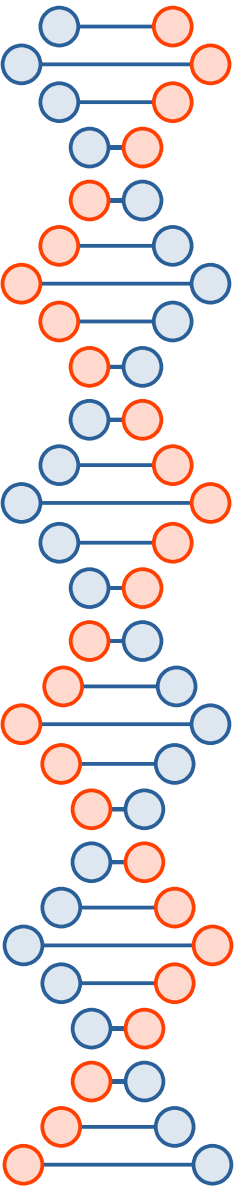
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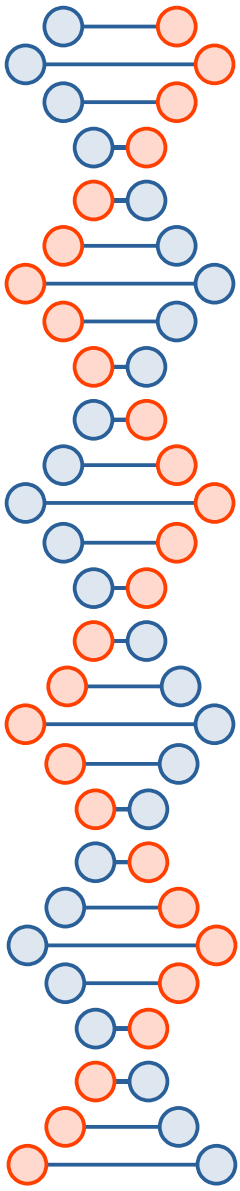
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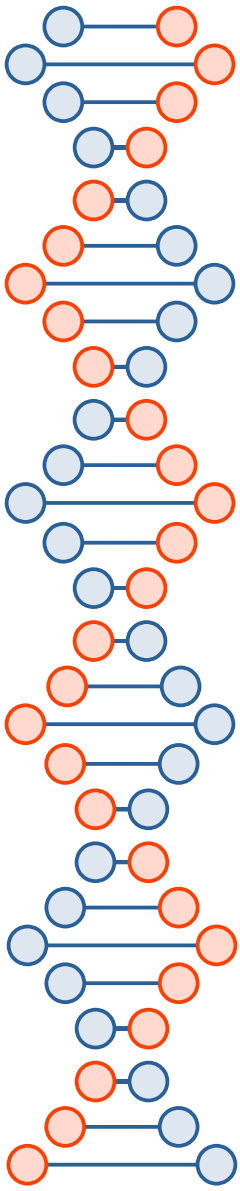


Example 3

Consider **$N = 100$** spherical amorphous silica particles suspended in a well-mixed water volume of **$V = 1\text{L}$** . Initially, the concentration of the dissolved silica (silicic acid) in the water is **$Si_{ini} = 1\text{ mM}$** and the radius of the particles is **$r_{ini} = 0.1\text{ }\mu\text{m}$** (the same for all particles). Assume the rate constant for silica dissolution of **$k_d = 1.6\text{ mol m}^{-2}\text{ yr}^{-1}$** and for silica precipitation of **$k_p = 0.8\text{ m yr}^{-1}$** .

The net rate of dissolution of silica particles is calculated according to $\frac{dSi}{dt} = -k_p \times \frac{A}{V} \times (Si - Si_{eq})$

where $A = N \times 4\pi r^2$ is the total surface area of the particles and $Si_{eq} = \frac{k_d}{k_p}$ is the equilibrium concentration.



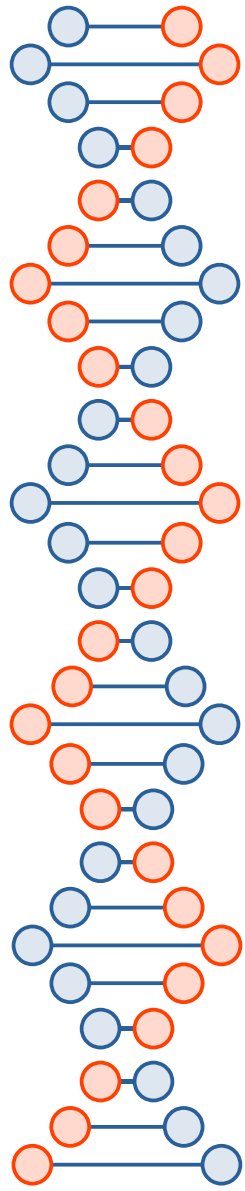
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What is the **initial rate** of increase in the concentration of silicic acid due to the dissolution of the silica particles?



Example 3

Consider **$N = 100$** spherical amorphous silica particles suspended in a well-mixed water volume of **$V = 1\text{L}$** . Initially, the concentration of the dissolved silica (silicic acid) in the water is **$Si_{ini} = 1\text{ mM}$** and the radius of the particles is **$r_{ini} = 0.1\text{ mm}$** (the same for all particles). Assume the rate constant for silica dissolution of **$k_d = 1.6\text{ mol m}^{-2}\text{ yr}^{-1}$** and for silica precipitation of **$k_p = 0.8\text{ m yr}^{-1}$** .

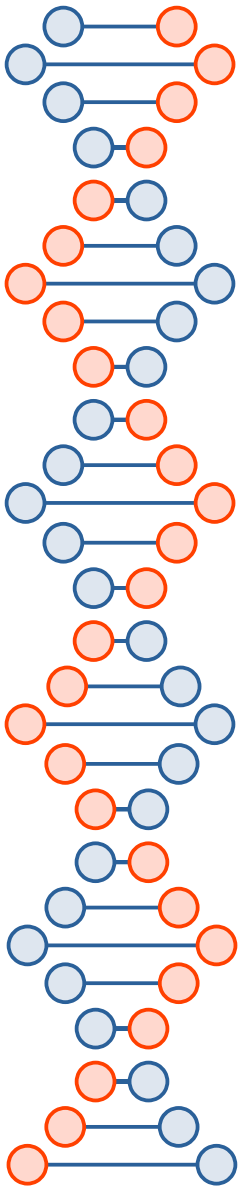
The net rate of dissolution of silica particles is calculated according to $\frac{dSi}{dt} = -k_p \times \frac{A}{V} \times (Si - Si_{eq})$

where $A = N \times 4\pi r^2$ is the total surface area of the particles and $Si_{eq} = \frac{k_d}{k_p}$ is the equilibrium concentration.

What is the **initial rate** of increase in the concentration of silicic acid due to the dissolution of the silica particles?

1. First, we evaluate the equilibrium and initial concentrations in SI units:

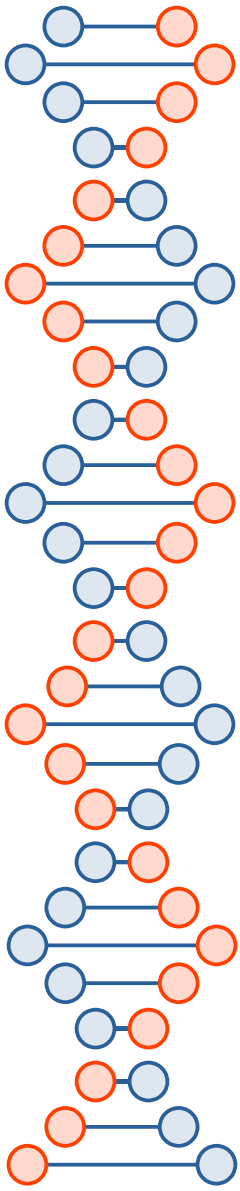
$$Si_{eq} = \frac{1.6\text{ mol m}^{-2}\text{ yr}^{-1}}{0.8\text{ m yr}^{-1}} = 2\frac{\text{mol m}^{-2}}{\text{m}} = 2\text{ mol m}^{-3} \quad Si_{ini} = 1\text{ mM} = 1 \times 10^{-3} \frac{\text{mol}}{\text{L}} = 1 \times 10^{-3} \frac{\text{mol}}{10^{-3}\text{ m}^3} = 1\text{ mol m}^{-3}$$



Example 3

2. Then, we substitute the values to the formula and follow the steps 1–3 described above:

$$\left(\frac{dSi}{dt}\right)_{ini} = -0.8 \frac{m}{yr} \times \frac{100 \times 4 \pi \times (0.1 mm)^2}{1 L} \times (1 mol m^{-3} - 2 mol m^{-3})$$

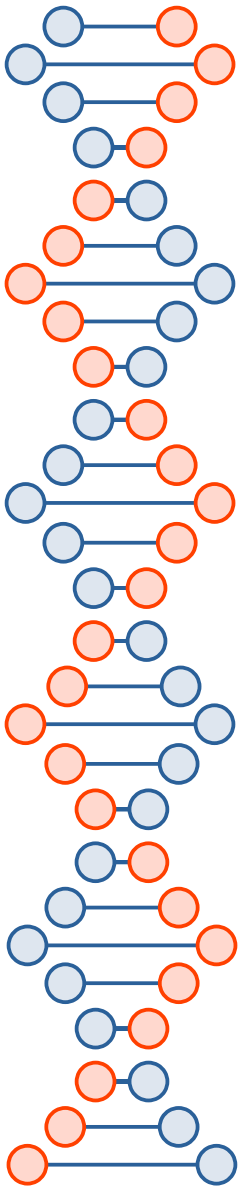


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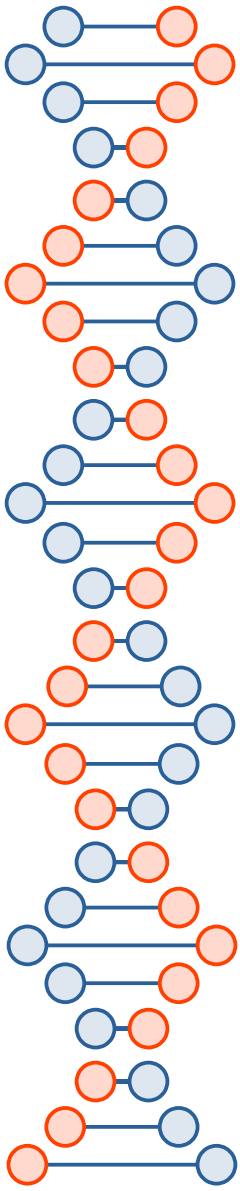
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$$= -0.8 \times 400 \pi \frac{m}{yr} \times \frac{10^{-2} \times 10^{-6} \text{ m}^2}{10^{-3} \text{ m}^3} \times (-1) \text{ mol m}^{-3}$$



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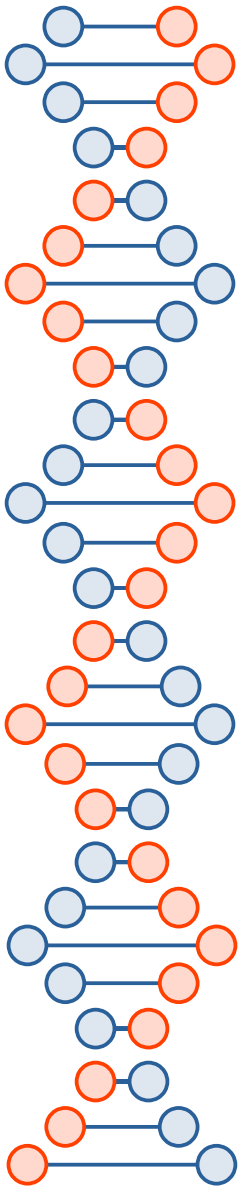
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$$\approx 1000 \frac{m}{yr} \times \frac{10^{-5}}{m} \text{ mol m}^{-3}$$



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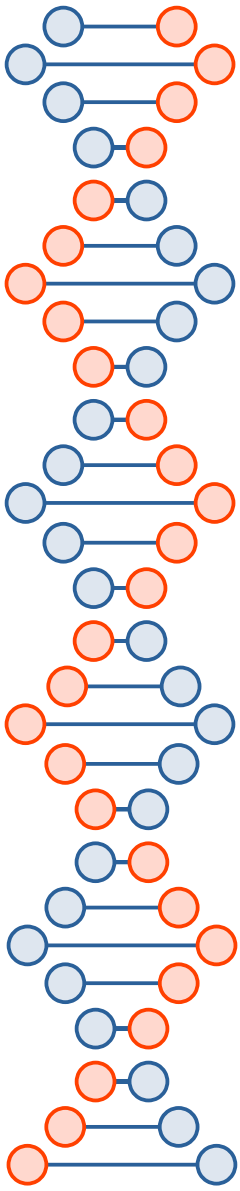
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$$\approx 1000 \frac{m}{yr} \times \frac{10^{-5}}{m} \text{ mol m}^{-3}$$

$$= 0.01 \text{ mol m}^{-3} \text{ yr}^{-1}$$



It *is* simple, isn't it?!

Just carefully follow these basic steps:

- 1: Convert all prefixes to the form **10^n** , where **n** is an **integer** number.
- 2: Convert all units to **base SI units**.
- 3: **Simplify the expression** with units using basic mathematical rules.

We hope that these tips will be useful.